Neural Layer Matrices and Activations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

[Matrices, Matrix Algebra & Inverses · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

2 **[Foundational]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Let A and B be invertible $n \times n$ matrices. Prove algebraically that: (a) $(AB)^T = B^T A^T$, and (b) $(AB)^{-1} = B^{-1}A^{-1}$.

(a) Transpose Proof: The (i, j)-entry of $(AB)^T$ is the (j, i)-entry of AB: $[(AB)^T]_{ij} = [AB]_{ji} = \sum_{k=1}^{n} A_{jk}B_{ki} = \sum_{k=1}^{n} B_{ki}A_{jk}$. Using the definition of matrix transpose, $B_{ki} = [B^T]_{ik}$ and $A_{jk} = [A^T]_{kj}$: $[(AB)^T]_{ij} = \sum_{k=1}^{n} [B^T]_{ik}[A^T]_{kj} = [B^T A^T]_{ij}$. Since entries match for all i, j, $(AB)^T = B^T A^T$. ■ (b) Inverse Proof: A matrix C is the inverse of AB if and only if $(AB)C = I_n$ and $C(AB) = I_n$. Test.

3 **[Foundational]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Determine, with formal verification via the Subspace Test, whether $W = \{A \in M_2(\mathbb{R}) : A^T = -A\}$ (the set of skew-symmetric matrices) is a subspace of $M_2(\mathbb{R})$. Find a basis and $\dim(W)$.

Subspace Test Verification:Non-emptiness: The zero matrix O satisfies $O^T = O = -O$, so $O \in W$ .Additivity: Let $A, B \in W$. Then $(A + B)^T = A^T + B^T = (-A) + (-B) = -(A + B)$. Thus $A + B \in W$.Scalar Closure: Let $A \in W$ and $c \in \mathbb{R}$. Then $(cA)^T = c(A^T) = c(-A) = -(cA)$. Thus $cA \in W$.Since all 3 axioms hold, W is a vector subspace of $M_2(\mathbb{R})$. General form and basis: A matrix $A = (a, b; c, d)$ satisfies.

4 **[Foundational]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

Express the invertible matrix $A = (1, 2; 3, 4)$ as a product of elementary matrices.

Step 1: Reduce A to I_2 via row operations: 1. $R_2 \to R_2 - 3R_1$: Elementary matrix $E_1 = (1, 0; -3, 1)$. Result: $(1, 2; 0, -2)$. 2. $R_2 \to -\frac{1}{2}R_2$: Elementary matrix $E_2 = (1, 0; 0, -1/2)$. Result: $(1, 2; 0, 1)$. 3. $R_1 \to R_1 - 2R_2$: Elementary matrix $E_3 = (1, -2; 0, 1)$. Result: I_2. Step 2: Invert elementary matrices: We have $E_3E_2E_1A = I_2 \Rightarrow A = E_1^{-1}E_2^{-1}E_3^{-1}$. Computin.

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Practice Problem 2: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = abla(x^3y^2 + \sin(xz))$ and C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \le t \le 2\pi$.

Since $\mathbf{F}$ is explicitly given as the gradient of the potential function $f(x, y, z) = x^3y^2 + \sin(xz)$, by the Fundamental Theorem of Line Integrals: $\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{r}(2\pi)) - f(\mathbf{r}(0))$. Initial point: $\mathbf{r}(0) = \langle \cos 0, \sin 0, 0 \rangle = (1, 0, 0)$. Terminal point: $\mathbf{r}(2\pi) = \langle \cos(2\pi), \sin(2\pi), 2\pi \rangle = (1, 0, 2\pi)$. Evaluating f at endpoints: $f(1, 0, 0) = 1^3(0)^2 + \sin(0) = 0$. .

6 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 3 · Advanced / University RigorSuppose an $n \times n$ matrix A satisfies the polynomial equation $A^3 - 4A^2 + 2A - 5I = 0$. Prove that A is invertible, and express A^{-1} as an explicit polynomial in A.

Step 1 · Isolate the Identity Matrix: $A^3 - 4A^2 + 2A = 5I$. Divide both sides by 5: $\frac{1}{5}(A^3 - 4A^2 + 2A) = I$. Step 2 · Factor out A: $A\left[\frac{1}{5}(A^2 - 4A + 2I)\right] = I$. Similarly, since polynomials in A commute: $\left[\frac{1}{5}(A^2 - 4A + 2I)\right]A = I$. Conclusion: By the definition of matrix inverse, A is invertible, and its inverse is: $\mathbf{A^{-1} = \frac{1}{5}A^2 - \frac{4}{5}A + \frac{2}{5}I}$.

7 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Practice Problem 2: Find the mass of a solid right circular cylinder of radius R and height h whose density at any point is proportional to its distance from the axis of symmetry, $\delta(r, \theta, z) = kr$ for constant $k > 0$.

Using cylindrical coordinates, the cylinder E is bounded by $0 \le r \le R$, $0 \le \theta \le 2\pi$, $0 \le z \le h$. The differential volume is $dV = r\,dz\,dr\,d\theta$. The total mass is: $M = \iiint_E \delta\,dV = \int_0^{2\pi}\int_0^R\int_0^h (kr) \cdot r\,dz\,dr\,d\theta = k\left(\int_0^{2\pi} d\theta\right)\left(\int_0^h dz\right)\left(\int_0^R r^2\,dr\right)$. $M = k(2\pi)(h)\left(\frac{R^3}{3}\right) = \frac{2\pi khR^3}{3}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Practice Problem 1: Evaluate $\iiint_E (x^2 + y^2 + z^2)\,dV$, where E is the solid ball $x^2 + y^2 + z^2 \le a^2$.

In spherical coordinates, $x^2 + y^2 + z^2 = \rho^2$ and $dV = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$. The ball E is defined by $0 \le \rho \le a$, $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$. $\iiint_E \rho^2 \cdot (\rho^2 \sin\phi)\,d\rho\,d\phi\,d\theta = \left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi} \sin\phi\,d\phi\right)\left(\int_0^a \rho^4\,d\rho\right)$. Evaluating each integral: $2\pi \times 2 \times \frac{a^5}{5} = \frac{4\pi a^5}{5}$.

© ClipSAT — clipsat.org | All Rights Reserved

9 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 1 · FoundationsLet $M_{2\times 2}$ be the vector space of all 2×2 real matrices. Show that the set of all symmetric matrices $W=\{A\in M_{2\times 2}: A^T=A\}$ is a subspace of $M_{2\times 2}$.

Step 1 · Zero vector: The 2×2 zero matrix satisfies $0^T=0$, so $0\in W$. Step 2 · Closed under addition: Let $A,B\in W$. Then $A^T=A$ and $B^T=B$. Using transpose properties: $(A+B)^T=A^T+B^T=A+B\Rightarrow A+B\in W$. Step 3 · Closed under scalar multiplication: Let $A\in W$ and $c\in\mathbb{R}$: $(cA)^T=cA^T=cA\Rightarrow cA\in W$. All three conditions hold, so W is a subspace of $M_{2\times 2}$.

10 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 2 · MediumLet A and B be 3×3 matrices with $\det(A)=3$ and $\det(B)=-2$. Compute the exact numerical value of: $\det(2A^2B^{-1}A^T)$.

Apply determinant algebraic properties for 3×3 matrices: $\det(2M)=2^3\det(M)=8\det(M)$. Using the multiplicative, power, transpose, and inverse properties: $\det(A^2B^{-1}A^T)=[\det(A)]^2\cdot\frac{1}{\det(B)}\cdot\det(A)=\frac{[\det(A)]^3}{\det(B)}$. Substitute $\det(A)=3$ and $\det(B)=-2$: $\det(2A^2B^{-1}A^T)=8\cdot\frac{3^3}{-2}=8\cdot\frac{27}{-2}=8(-13.5)=\mathbf{-108}$.

11 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 2 · MediumLet $A=(3,5;1,2)$ and $\mathbf{b}=(4;1)$. Compute A^{-1} and use it to solve the linear system $A\mathbf{x}=\mathbf{b}$.

Step 1 · Invert A: $\det(A)=(3)(2)-(5)(1)=6-5=1$. $A^{-1}=\frac{1}{1}(2,\ -5;\ -1,3)=(2,\ -5;\ -1,3)$. Step 2 · Multiply $\mathbf{x}=A^{-1}\mathbf{b}$: $\mathbf{x}=(2,\ -5;\ -1,3)(4;1)=(2(4)-5(1);\ -1(4)+3(1))=(\mathbf{3;\ -1})$.

12 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Level 1 · FoundationsLet $A=(2,\ -1,3;4,0,\ -2)$ and $\mathbf{x}=(3;\ -2;1)$. Compute $A\mathbf{x}$ explicitly using the linear combination of columns definition.

Using the column combination perspective: $A\mathbf{x}=3(2;4)+(-2)(-1;0)+1(3;\ -2)=(6;12)+(2;0)+(3;\ -2)=(\mathbf{11;10})$.

13 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

[Determinants & Geometric Volume · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10)=6(10)+6=\mathbf{66}$. Setting $6t+6=72$ gives $6t=66\Rightarrow t=\mathbf{11}$.

14 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

[Eigenvalues, Eigenvectors & Diagonalization · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10)=6(10)+6=\mathbf{66}$. Setting $6t+6=72$ gives $6t=66\Rightarrow t=\mathbf{11}$.

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization**

[Systems of Linear Equations & Row Reduction · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

16 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

[Vector Spaces, Subspaces & The Four Fundamental Subspaces · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

17 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

[Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS**

[Applications of Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification**

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 **[Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

[Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{10}$ to $\mathbf{y} \in \mathbb{R}^{5}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. **55** ✓

b. 50

c. 15

d. 65

The weight matrix $\mathbf{W}$ has shape 5×10, giving 50 weights. The bias vector $\mathbf{b}$ has 5 parameters. Total = $50 + 5 = 55$.

22 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -2$ and $z_2 = 3$.

a. $f(z_1) = 2, \; f(z_2) = 3$

b. $f(z_1) = -2, \; f(z_2) = 3$

c. $f(z_1) = 0, \; f(z_2) = 0$

d. **$f(z_1) = 0, \; f(z_2) = 3$** ✓

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \le 0$. Thus $f(-2) = 0$ and $f(3) = 3$.

23 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a linear layer with a weight matrix W and an input vector $\mathbf{x}$. If $W = (1, 2; 3, 4)$ and $\mathbf{x} = (5; 6)$, compute the output vector $\mathbf{y} = W\mathbf{x}$.

a. $(17; 39)$ ✓

b. $(11; 19)$

c. $(17; 19)$

d. $(11; 39)$

The output vector $\mathbf{y}$ is computed by multiplying the weight matrix W by the input vector $\mathbf{x}$. This is a standard matrix-vector multiplication. $\\ \\ \mathbf{y} = W\mathbf{x} = (1, 2; 3, 4)(5; 6)\\\ \\$ To compute the first element of $\mathbf{y}$, we take the dot product of the first row of W with $\mathbf{x}$: $(1)(5) + (2)(6) = 5 + 12 = 17.\\ \\$ To compute the second element of $\mathbf{y}$, we take the dot product of the second row of W with $\mathbf{x}$.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a linear transformation represented by the matrix $M = [2, k; 1, 3]$. A vector $\mathbf{v} = [1; 2]$ is transformed by M into $\mathbf{v}' = M\mathbf{v} = [8; 7]$. Determine the value of the unknown parameter k.

a. 1

b. **2** ✓

c. 3

d. 4

The problem involves a matrix-vector multiplication. We are given the matrix $M = [2, k; 1, 3]$ and the vectors $\mathbf{v} = [1; 2]$ and $\mathbf{v}' = [8; 7]$. The transformation is defined by $\mathbf{v}' = M\mathbf{v}$. Let's perform the matrix-vector multiplication: $[8; 7] = [2, k; 1, 3][1; 2]$ This expands to two equations: 1. $8 = (2)(1) + (k)(2)$ 2. $7 = (1)(1) + (3)(2)$ Let's solve the first equation for k: $8 = 2 + 2k$ Subtract 2.

25 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A linear transformation is defined by the matrix $M = (2, 1; 3, 4)$. If an input vector is $v = (x; 1)$, and the output vector after the transformation is $w = (5; 7)$, what is the value of x?

a. **1** ✓

b. 2

c. 3

d. 4

The linear transformation is given by $w = Mv$. Substituting the given matrices and vectors, we have: $(3; 7) = (2, 1; 3, 4)(x; 1)$ Performing the matrix multiplication on the right side gives: $(3; 7) = (2x + 1(1); 3x + 4(1)) = (2x + 1; 3x + 4)$ Equating the corresponding components of the vectors, we get two equations: 1. $3 = 2x + 1$ 2. $7 = 3x + 4$ Solving the first equation for x: $3 - 1 = 2x$ $2 = 2x$ $x = \frac{2}{2}$ $x = 1$ Solving.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a 2D dataset represented by the points $A = (1, 2)$, $B = (4, 2)$, and $C = (2, 5)$. If this dataset undergoes a linear transformation represented by the matrix $M = (2, 0; 0, 1)$, what are the coordinates of the transformed points A', B', and C'?

a. $A' = (2, 2), B' = (8, 2), C' = (4, 5)$ ✓

b. $A' = (1, 4), B' = (4, 4), C' = (2, 10)$

c. $A' = (2, 4), B' = (8, 4), C' = (4, 10)$

d. $A' = (3, 2), B' = (6, 2), C' = (4, 7)$

The transformation of a point $\mathbf{x}$ by a matrix M is given by $M\mathbf{x}$. We apply the matrix $M = (2, 0; 0, 1)$ to each point: For point $A = (1, 2)$, the transformed point A' is:
$(2, 0; 0, 1)(1; 2) = (2 \times 1 + 0 \times 2; 0 \times 1 + 1 \times 2) = (2; 2)$ So, $A' = (2, 2)$. For point $B = (4, 2)$, the transformed point B' is: $(2, 0; 0, 1)(4; 2) = (2; 2)$ So, $B' = (8, 2)$. For point.

27 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a dataset with n data points $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, where each $\mathbf{x}_i \in \mathbb{R}^p$. The sample covariance matrix is given by $\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$, where $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i$ is the sample mean. Suppose PCA is performed on this dataset, resulting in principal components defined by the eigenvectors of $\mathbf{S}$ and corresponding variances given by the eigenvalues. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ be the orthonormal eigenvectors of $\mathbf{S}$ with associated eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$. The first principal component captures the direction of maximum variance, λ_1, and the second principal component captures the next largest variance, λ_2, orthogonal to the first. If we decide to reduce the dimensionality from p to k by keeping only the first k principal components, the proportion of the total variance retained is given by $\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{p} \lambda_i}$. Suppose for a particular dataset, the eigenvalues of the covariance matrix are $\{10.5, 5.2, 2.1, 0.8, 0.4\}$. If we aim to retain at least 90% of the total variance, what is the minimum number of principal components required?

a. **3** ✓

b. 2

c. 4

d. 5

The total variance in the dataset is the sum of all eigenvalues:
$\sum_{i=1}^{p} \lambda_i = 10.5 + 5.2 + 2.1 + 0.8 + 0.4 = 19.0$. We need to find the minimum k such that $\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{p} \lambda_i} \geq 0.90$.
Let's check for different values of k: For $k = 1$: The variance retained is $\frac{10.5}{19.0} \approx 0.553$, which is less than 0.90. For $k = 2$: The variance retained is $\frac{10.5 + 5.2}{19.0} = \frac{15.7}{19.0} \approx 0.826$, which is less than 0.90. For $k = 3$: The variance.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A binary classification model uses a logistic sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is defined by $\sigma(z) = 0.5$, where $z = w_1x_1 + w_0$. For a dataset with two features x_1 and x_2, the model's linear component is $z = 2x_1 - x_2 + b$. If the classification boundary passes through the point $(x_1, x_2) = (3, 1)$, what is the value of the bias b?

a. $b = -5$

b. $b = -7$ ✓

c. $b = 5$

d. $b = 7$

The logistic sigmoid function is $\sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is where $\sigma(z) = 0.5$. Setting $\sigma(z) = 0.5$ gives $\frac{1}{1+e^{-z}} = 0.5$, which implies $1 = 0.5(1 + e^{-z})$, so $2 = 1 + e^{-z}$, and $e^{-z} = 1$. This occurs when $-z = 0$, or $z = 0$. Therefore, the decision boundary is defined by $z = 0$. The linear component of the model is given by $z = w_1x_1 + w_2x_2 + b$. In this problem, $z = 2x_1 - x_2 + b$. Since the decision boundary is $z = 0$, we have.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a dataset of observations $\{x_i\}_{i=1}^{n}$ where each x_i is a 2-dimensional vector. The sample covariance matrix for this dataset is given by: $S = (4,\ -2;\ -2, 1)$ Principal Component Analysis (PCA) is applied to this dataset. The first principal component corresponds to the direction with the largest variance. Let the direction vector of the first principal component be $\mathbf{v}_1 = (a; b)$. The variance explained by this principal component is given by the corresponding eigenvalue λ_1. If $\mathbf{v}_1$ is a unit vector, what is the value of $a^2 + b^2$?

a. **1** ✓

b. 2

c. 4

d. 5

The first principal component direction vector $\mathbf{v}_1$ is an eigenvector of the covariance matrix S. The variance explained by this component is the corresponding eigenvalue λ_1. The problem states that $\mathbf{v}_1 = (a; b)$ is a unit vector. By definition, a unit vector has a magnitude of 1. The squared magnitude of a vector $(a; b)$ is $a^2 + b^2$. Therefore, if $\mathbf{v}_1$ is a unit vector, its squared magnitude must be 1, which means $a^2 + b^2 = 1$. The specific va.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a dataset with two features, X_1 and X_2. The covariance matrix for this dataset is given by: $\Sigma = (10, 5; 5, 20)$ What is the variance of the first feature, X_1?

a. **10** ✓

b. 5

c. 20

d. 25

The covariance matrix Σ for a dataset with two features X_1 and X_2 is structured as follows: $\Sigma = (Cov(X_1, X_1), Cov(X_1, X_2); Cov(X_2, X_1), Cov(X_2, X_2))$ By definition, the variance of a random variable is the covariance of the variable with itself. Therefore, $Var(X_1) = Cov(X_1, X_1)$ and $Var(X_2) = Cov(X_2, X_2)$. In the given covariance matrix: $\Sigma = (10, 5; 5, 20)$ The element in the first row and first column represents $Cov(X_1, X_1)$, which is the variance.

31 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a simple logistic regression model where the decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. For a data point (x_1, x_2), if $-1.5x_1 + 2x_2 + 5 < 0$, it is classified into Class 0. Otherwise, it is classified into Class 1. Which of the following points lies on the decision boundary?

a. $(-1, 1)$

b. $(2, -2)$

c. $(0, -2.5)$ ✓

d. $(3, 0.5)$

The decision boundary is defined by the equation $-1.5x_1 + 2x_2 + 5 = 0$. To find which point lies on the boundary, we substitute the coordinates of each point into the equation. For point $(-1, 1)$: $-1.5(-1) + 2(1) + 5 = 1.5 + 2 + 5 = 8.5$, which is not 0. For point $(2, -2)$: $-1.5(2) + 2(-2) + 5 = -3 - 4 + 5 = -2$, which is not 0. For point $(0, -2.5)$: $-1.5(0) + 2(-2.5) + 5 = 0 - 5 + 5 = 0$. This point lies on the boundary. For point $(3, 0.5)$: $-1.5(3) + 2(0.5) + 5 = -4.5 + 1 + 5 = 1.5$, which is not 0. Th.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a simple linear regression model where the predicted value $\hat{y}$ is given by $\hat{y} = b_0 + b_1 x$. If we have two data points $(x_1, y_1) = (2, 5)$ and $(x_2, y_2) = (4, 9)$, and the Ordinary Least Squares (OLS) method yields the regression line $\hat{y} = 1 + 2x$, what is the predicted value $\hat{y}$ for $x = 3$?

a. 5

b. 6

c. 7 ✓

d. 8

The problem provides a simple linear regression equation $\hat{y} = b_0 + b_1 x$ and specific values for the intercept $b_0 = 1$ and the slope $b_1 = 2$. We are asked to find the predicted value $\hat{y}$ for a given value of x. We substitute $x = 3$ into the regression equation: $\hat{y} = 1 + 2(3)$ $\hat{y} = 1 + 6$ $\hat{y} = 7$ Therefore, the predicted value $\hat{y}$ for $x = 3$ is 7. The data points $(2, 5)$ and $(4, 9)$ are consistent with this regression line: for $x = 2$, $\hat{y} = 1 + 2(2) = 5$, and for.

33 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider the quadratic loss function $L(w) = w^2 - 4w + 3$. Find the value of w where the function reaches its minimum.

a. $w = 1$

b. $w = 2$ ✓

c. $w = 3$

d. $w = 4$

The given loss function is a quadratic function $L(w) = w^2 - 4w + 3$. This represents a parabola opening upwards. The minimum value of a quadratic function of the form $aw^2 + bw + c$ occurs at $w = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -4$. Therefore, the minimum occurs at $w = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Alternatively, we can find the minimum by taking the derivative and setting it to zero: $L'(w) = 2w - 4$. Setting $L'(w) = 0$ gives $2w - 4 = 0$, which implies $2w = 4$, so $w = 2$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a simple logistic regression model where the decision boundary is defined by the linear equation $w_1x_1 + w_2x_2 + b = 0$. If the weights are $w_1 = 2$ and $w_2 = 3$, and the bias is $b = -7$, what is the equation of the decision boundary?

a. $2x_1 + 3x_2 - 7 = 0$ ✓

b. $3x_1 + 2x_2 + 7 = 0$

c. $2x_1 + 3x_2 + 7 = 0$

d. $-2x_1 - 3x_2 + 7 = 0$

The decision boundary in a logistic regression model is defined by the equation where the log-odds are zero. This corresponds to a linear combination of the input features and weights, plus the bias, being equal to zero. Given the weights $w_1 = 2$ and $w_2 = 3$, and the bias $b = -7$, the equation of the decision boundary is $w_1x_1 + w_2x_2 + b = 0$. Substituting the given values, we get $2x_1 + 3x_2 + (-7) = 0$, which simplifies to $2x_1 + 3x_2 - 7 = 0$.

35 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider the following quadratic loss function: $L(w) = 3w^2 - 12w + 10$. What is the value of w that minimizes this loss function?

a. **2** ✓

b. 1

c. 4

d. 12

To find the minimum of the quadratic loss function $L(w) = 3w^2 - 12w + 10$, we can find the derivative with respect to w and set it to zero. The derivative of $L(w)$ with respect to w is: $\frac{dL}{dw} = 6w - 12$. Setting the derivative to zero to find the critical point: $6w - 12 = 0$ $6w = 12$ $w = \frac{12}{6}$ $w = 2$. The second derivative is $\frac{d^2L}{dw^2} = 6$, which is positive, confirming that $w = 2$ corresponds to a minimum.

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a simple linear regression model where a dependent variable y is predicted from an independent variable x. The model is given by the equation $y = \beta_0 + \beta_1 x + \varepsilon$, where ε is the error term. Suppose we have estimated the regression line to be $\hat{y} = 3 + 2x$. If we observe an independent variable value of $x = 4$, what is the predicted value $\hat{y}$ for the dependent variable?

a. **11** ✓

b. 14

c. 10

d. 7

The problem asks for the predicted value $\hat{y}$ given an independent variable value $x = 4$ and a linear regression equation $\hat{y} = 3 + 2x$. We substitute the value of x into the equation: $\hat{y} = 3 + 2(4)$ $\hat{y} = 3 + 8$ $\hat{y} = 11$ Therefore, the predicted value for the dependent variable is 11. Common errors might involve incorrectly multiplying 2 by 4 or incorrectly adding 3 to 8.

37 [Intermediate] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a simple quadratic loss function $L(w) = (w^2 - 4w + 5)$. We want to find the minimum of this loss function using gradient descent. The gradient of the loss function with respect to w is given by $\frac{dL}{dw} = 2w - 4$. If we start at an initial weight $w_0 = 5$ and use a learning rate $\alpha = 0.5$, what will be the value of the weight w_1 after the first update step, i.e., $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$?

a. 4 ✓

b. 3

c. 4.5

d. 5

The loss function is $L(w) = w^2 - 4w + 5$. The gradient is $\frac{dL}{dw} = 2w - 4$. We are given an initial weight $w_0 = 5$ and a learning rate $\alpha = 0.5$. The gradient at w_0 is $\frac{dL}{dw}(w_0) = 2(5) - 4 = 10 - 4 = 6$. The update rule for gradient descent is $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$. Substituting the values, we get $w_1 = 5 - 0.5 \times 6 = 5 - 3 = 2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Consider a simple linear regression model where the predicted value $\hat{y}$ is related to the independent variable x by the equation $\hat{y} = mx + c$. If the line of best fit passes through the points $(1, 5)$ and $(3, 11)$, what is the slope m of this line?

a. **3** ✓

b. 2

c. 5

d. 6

The slope m of a line passing through two points (x_1, y_1) and (x_2, y_2) is given by the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Given the points $(1, 5)$ and $(3, 11)$, we have $x_1 = 1$, $y_1 = 5$, $x_2 = 3$, and $y_2 = 11$. Plugging these values into the formula, we get $m = \frac{11 - 5}{3 - 1} = \frac{6}{2} = 3$. Therefore, the slope of the line of best fit is 3.

39 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Consider a discrete random variable X with the following probability mass function (PMF): $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. What is the value of $P(X \leq 2)$?

a. $\frac{3}{10}$ ✓

b. $\frac{1}{10}$

c. $\frac{6}{10}$

d. $\frac{4}{10}$

The probability mass function is given by $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. We need to find $P(X \leq 2)$. This is the sum of probabilities for $X = 1$ and $X = 2$. $P(X \leq 2) = P(X = 1) + P(X = 2)$. $P(X = 1) = \frac{1}{10}$ and $P(X = 2) = \frac{2}{10}$. Therefore, $P(X \leq 2) = \frac{1}{10} + \frac{2}{10} = \frac{3}{10}$.

40 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider a dataset where the number of defective items produced by a machine per hour, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The probability mass function (PMF) for a Poisson distribution is given by $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$. What is the value of λ in this scenario?

a. $\lambda = 0$

b. $\lambda = 1$

c. $\lambda = 3$ ✓

d. $\lambda = e^{-3}$

The problem statement explicitly defines that the number of defective items, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The rate parameter λ is a direct characteristic of the Poisson distribution that determines its mean and variance. Therefore, the value of λ is directly given as 3.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider the following quadratic loss function for a regression problem: $L(y, \hat{y}) = (y - \hat{y})^2$. If the predicted value $\hat{y} = 5$ and the true value $y = 3$, what is the value of the loss function?

a. 4 ✓

b. -2

c. 9

d. 2

The quadratic loss function is defined as $L(y, \hat{y}) = (y - \hat{y})^2$. Given the true value $y = 3$ and the predicted value $\hat{y} = 5$, we substitute these values into the loss function: $L(3, 5) = (3 - 5)^2$ $L(3, 5) = (-2)^2$ $L(3, 5) = 4$ Therefore, the value of the loss function is 4.

42 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider two events, A and B, with the following probabilities: $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. What is the conditional probability of A given B, denoted as $P(A|B)$?

a. 0.2

b. **0.4** ✓

c. 0.5

d. 0.8

The formula for conditional probability is $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Given $P(A \cap B) = 0.2$ and $P(B) = 0.5$, we can substitute these values into the formula: $P(A|B) = \frac{0.2}{0.5} = 0.4$. Therefore, the conditional probability of A given B is 0.4.

43 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

A centered feature matrix $\mathbf{X}$ has shape $N \times 14$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

a. 14×14 **symmetric matrix** ✓

b. $N \times N$ symmetric matrix

c. 14×14 skew-symmetric matrix

d. $N \times 14$ rectangular matrix

Since $\mathbf{X}$ has 14 columns, $\mathbf{X}^T$ is $14 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 14×14. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider a 2D data point $\mathbf{v} = [3;\ -4]$. What is the squared Euclidean norm (or squared magnitude) of this vector?

a. 7

b. **25** ✓

c. 9

d. 16

The squared Euclidean norm of a vector $\mathbf{v} = [v_1; v_2]$ is given by $||\mathbf{v}||^2 = v_1^2 + v_2^2$. For the given vector $\mathbf{v} = [3;\ -4]$, the squared Euclidean norm is $(3)^2 + (-4)^2 = 9 + 16 = 25$.

45 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Given $n = 4$ independent observations from a normal distribution with sum $\sum_{i=1}^{4} x_i = 12$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?

a. $\hat{\mu}_{\text{MLE}} = 3$ ✓

b. $\hat{\mu}_{\text{MLE}} = 6$

c. $\hat{\mu}_{\text{MLE}} = 12$

d. $\hat{\mu}_{\text{MLE}} = 1.5$

The log-likelihood $\ln L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\frac{\partial \ln L}{\partial \mu} = 0$ yields the sample mean $\hat{\mu} = \frac{1}{n}\sum x_i = \frac{12}{4} = 3$.

© ClipSAT — clipsat.org | All Rights Reserved